

Izpitna naloga (projekt)

Predmet: Računalništvo v oblaku in mobilne rešitve

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Nosilec predmeta: Dušan Fugina

Namestitev Wordpress-a na EC2 instanco (AWS)

Opis projekta

Cilj projekta

Glavni cilj projekta je bil namestiti WordPress na AWS EC2 instanco, da bi imeli zagotovljeno hitrejše delovanje in večjo skabilnost spletne strani v primerjavi s klasičnimi strežniki. Ključna prednost te rešitve je popoln nadzor nad viri, saj nam EC2 omogoča, da kadarkoli povečamo procesorsko moč, delovni pomnilnik ali prostor na disku glede na rast prometa. Z uporabo namenske infrastrukture in optimiziranega spletnega strežnika Nginx smo dosegli krajše odzivne čase, hkrati pa s prilagodljivimi varnostnimi nastavitvami poskrbeli za višjo stopnjo zaščite podatkov in boljšo uporabniško izkušnjo.

Izvedba projekta

1. Najprej sem ustvaril Virtual Private Cloud (VPC) z vsemi potrebnimi komponentami, vključno s podomrežjem (subnet), usmerjevalno tabelo (route table) in internetnim prehodom (internet gateway), kar je omogočilo javni dostop do instance. Nato sem vzpostavil EC2 instanco z izbranim operacijskim sistemom Debian, pri čemer sem natančno definiral parametre, kot so tip instance, velikost diska in omrežne nastavitve. Za varnost sem ustvaril varnostno skupino, ki dovoljuje promet na portih 22 (SSH) za moj IP, 80 (HTTP) in 443 (HTTPS) za vse IP (javni dostop), ter generiral ključ (key pair) za varno povezavo prek PuTTY.
2. Ko je bila EC2 instance uspešno zagnana, sem se prek PuTTY povezal nanjo z uporabo generiranega zasebnega ključa. Po vzpostavitvi povezave sem najprej posodobil seznam paketov z ukazom **sudo apt update** in nato nadgradil vse obstoječe pakete z ukazom **sudo apt upgrade**, s čimer sem zagotovil, da ima operacijski sistem nameščene najnovejše varnostne popravke in izboljšave.
3. Nato sem namestil spletni strežnik Nginx z ukazom **sudo apt install nginx**. Po namestitvi sem strežnik zagnal s **sudo systemctl start nginx** in ga omogočil, da se ob vsakem zagonu sistema samodejno zažene s **sudo systemctl enable nginx**. Sledila je konfiguracija v datoteki **/etc/nginx/sites-available/default**, kjer sem prilagodil nastavitve tako, da strežnik podpira WordPress, nastavljal sem mapo na **/var/www/html**, dodal podporo za PHP prek PHP-FPM ter konfiguriral pravila za pravilno usmerjanje zahtev. Dodatno sem prilagodil

še glavno konfiguracijsko datoteko Nginx (/etc/nginx/nginx.conf) za boljše delovanje in varnost, nato pa sem storitev znova naložil s **sudo systemctl restart nginx**.

4. Za podatkovno bazo sem namestil odprtokodno MariaDB z ukazom **sudo apt install mariadb-server**. Po namestitvi sem zagnal storitev s **sudo systemctl start mariadb** in jo omogočil ob zagonu sistema s **sudo systemctl enable mariadb**. Nato sem izvedel varnostno konfiguracijo nastavil sem root geslo za bazo, onemogočil oddaljen root dostop, odstranil anonimne uporabnike in testne baze ter znova naložil pravice dostopa. S tem sem zagotovil, da je baza pripravljena za varno uporabo s WordPressom.
5. Sledila je namestitvev PHP-ja in potrebnih razširitev, ki jih WordPress zahteva za nemoteno delovanje. Z ukazom **sudo apt install php php-fpm php-curl php-cli php-zip php-mysql php-xml php-mbstring php-gd php-xmllrpc php-imagick php-intl php-soap** sem namestil PHP ter module za povezovanje z bazo MariaDB, urejanje slik, delo s XML in JSON, podporo za več zlogov (mbstring) ter druge ključne funkcionalnosti. Po namestitvi sem zagnal storitev PHP-FPM in jo omogočil ob zagonu sistema.
6. Za enostavnejše upravljanje podatkovne baze sem namestil **phpMyAdmin**. Ker repozitoriji Debian ne vsebujejo vedno najnovejše različice, sem ga ročno prenesel z uradne spletne strani z ukazom **sudo wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz**. Nato sem ustvaril mapo za namestitev in razpakirano vsebino premaknil vanjo, ter nastavil ustrezne pravice dostopa.
7. Nazadnje sem namestil še WordPress. Z ukazom **sudo wget https://wordpress.org/latest.tar.gz** sem prenesel najnovejšo različico, jo razpakiral z **sudo tar -xzf latest.tar.gz** in nato vsebino razpakirane mape WordPress prestavil v mapo spletnega strežnika **/var/www/html** z ukazom **sudo mv wordpress/* /var/www/html/**. Povezal sem se še na PhpMyAdmin preko brskalnika in dodal novo podatkovno bazo z imenom **wordpressmarko**. Po teh korakih sem lahko v spletnem brskalniku odprl naslov EC2 instance in sledil čarovniku za namestitev WordPressa, kjer sem vnesel podatke za povezavo z bazo MariaDB, določil admin račun, inštaliral poljubno brezplačno temo in končal namestitev.

Zaslonske slike izvedbe

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EC2

Instances

Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Wordpress Marko

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Debian 13 (HVM), SSD Volume Type

ami-0da1f66573556d917 (64-bit (x86)) / ami-08241d277446b81d7 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Debian 13 (HVM), EBS General Purpose (SSD) Volume Type. Community developed free GNU/Linux distribution. <https://www.debian.org/>

Summary

Number of instances

1

Software Image (AMI)

Debian 13 (20260316-2418)

ami-0da1f66573556d917

Virtual server type (instance type)

t3.small

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 50 GiB

Cancel

Launch instance

Preview code

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EC2 > Instances > Launch an instance

Description

Debian 13 (HVM), EBS General Purpose (SSD) Volume Type. Community developed free GNU/Linux distribution.
https://www.debian.org/

Debian 13 (20260316-2418)

Architecture

64-bit...

AMI ID

ami-Oda1f66573556d917

Publish Date

2026-03-16

Username

admin

Verified provider

▼ Instance type

Info | Get advice

Instance type

t3.small

Family: t3 2 vCPU 2 GiB Memory Current generation: true

On-Demand SUSE base pricing: 0.055 USD per Hour

On-Demand RHEL base pricing: 0.0528 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0275 USD per Hour

On-Demand Linux base pricing: 0.024 USD per Hour

On-Demand Windows base pricing: 0.0424 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Wordpress kljuc

Create new key pair

▼ Network settings

Info

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Debian 13 (20260316-2418)

ami-Oda1f66573556d917

Virtual server type (instance type)

t3.small

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1 volume(s) - 50 GiB

Cancel

Launch instance

Preview code

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EC2 > Instances > Launch an instance

markotojagic

markotojagic

▼ Security group rule 1 (TCP, 22, 31.15.198.217/32)

Type

ssh

Protocol

TCP

Port range

22

Source type

My IP

Name

31.15.198.217/32

Description - optional

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Type

HTTP

Protocol

TCP

Port range

80

Source type

Anywhere

Source

0.0.0.0/0

Description - optional

e.g. SSH for admin desktop

Remove

▼ Security group rule 3 (TCP, 443)

Type

HTTPS

Protocol

TCP

Port range

443

Source type

Custom

Source

Description - optional

e.g. SSH for admin desktop

Remove

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

Add security group rule

▼ Summary

Number of instances

1

Software Image (AMI)

Debian 13 (20260316-2418)

ami-0da1f66573556d917

Virtual server type (instance type)

t3.small

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 50 GiB

Cancel

Launch instance

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EC2 > Instances > Launch an instance

Source type

Info

Source

Info

Description - optional

Info

Custom

Q Add CIDR, prefix list or securi

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Advanced network configuration

Configure storage

Info

Advanced

1x

50

GiB

gp3

Root volume, 16000 IOPS, Encrypted

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

S3 Files - new

EFS

FSx

Advanced details

Info

Summary

Number of instances

Info

1

Software Image (AMI)

Debian 13 (20260316-2418)

ami-Oda1f66573556d917

Virtual server type (instance type)

t3.small

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 50 GiB

Cancel

Launch instance

Preview code

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VPC > Your VPCs > vpc-01a5aa28ef1269995

VPC dashboard <

AWS Global View t2

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

vpc-01a5aa28ef1269995 / vpc_default

Actions

Details

Info

VPC ID

vpc-01a5aa28ef1269995

State

Available

DNS resolution

Enabled

Tenancy

default

Main network ACL

acl-09ab00e2db2ac1c69

Default VPC

No

IPV6 CIDR (Network border group)

-

Network Address Usage metrics

Disabled

Encryption control ID

-

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-01bc0d8807d0a0de4

IPV4 CIDR

10.0.0.0/24

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-078dafb03e445a35a

IPv6 pool

-

Owner ID

161457898876

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

VPC

Your AWS virtual network

vpc_default

Subnets (1)

Subnets within this VPC

eu-central-1a

subnet-wp

Route tables (1)

Route network traffic to resources

rtb-078dafb03e445a35a

Network Connections (1)

Connections to other networks

spletWP

CloudShell

Feedback

Console Mobile App

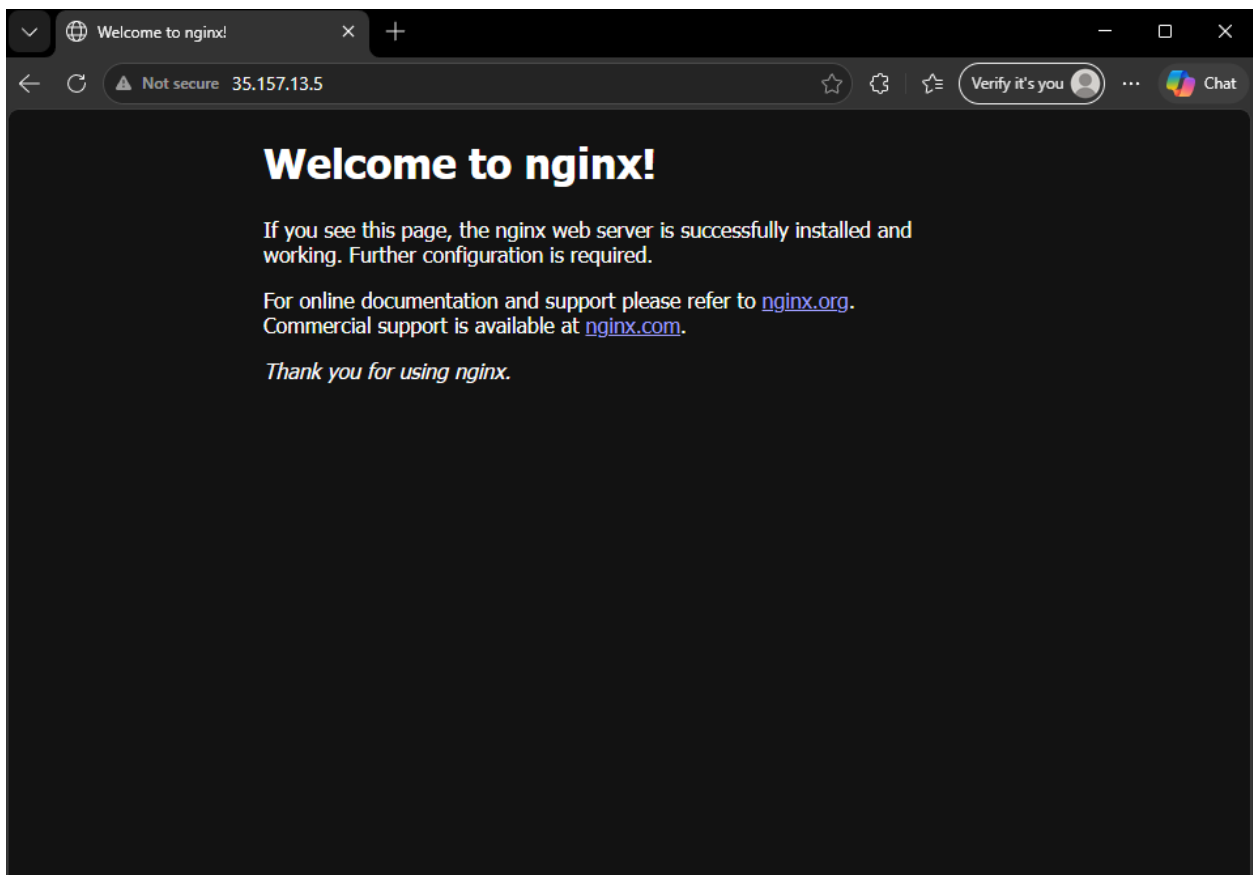
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```
admin@ip-10-0-0-150: ~  
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.  
Need to get 718 kB of archives.  
After this operation, 1891 kB of additional disk space will be used.  
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]  
Get:2 https://cdn-aws.deb.debian.org/debian trixie/main amd64 nginx-common all 1.26.3-3+deb13u2 [109 kB]  
Get:3 https://cdn-aws.deb.debian.org/debian trixie/main amd64 nginx amd64 1.26.3-3+deb13u2 [609 kB]  
Fetched 718 kB in 0s (10.5 MB/s)  
Preconfiguring packages ...  
Selecting previously unselected package nginx-common.  
(Reading database ... 31607 files and directories currently installed.)  
Preparing to unpack .../nginx-common_1.26.3-3+deb13u2_all.deb ...  
Unpacking nginx-common (1.26.3-3+deb13u2) ...  
Selecting previously unselected package nginx.  
Preparing to unpack .../nginx_1.26.3-3+deb13u2_amd64.deb ...  
Unpacking nginx (1.26.3-3+deb13u2) ...  
Setting up nginx-common (1.26.3-3+deb13u2) ...  
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.  
Setting up nginx (1.26.3-3+deb13u2) ...  
Upgrading binary: nginx.  
Processing triggers for man-db (2.13.1-1) ...  
admin@ip-10-0-0-150: ~$
```



```

GNU nano 8.4 /etc/nginx/sites-available/default
server {
    listen 80;

    root /var/www/html;
    index index.php index.html index.htm;
    client_max_body_size 8000M;

    location / {
        # try_files $uri $uri/ =404;
        try_files $uri $uri/ /index.php?$args;
    }

    # Pass PHP requests to PHP_FPM

    location ~ \.php$ {
        try_files $uri =404;
        fastcgi_buffer_size 256k;
        fastcgi_buffers 4 256k;
        fastcgi_busy_buffers_size 256k;

        fastcgi_split_path_info ^(.+\.php) (/.+)$;
        fastcgi_pass unix:/var/run/php/php8.4-fpm.sock;
        fastcgi_index index.php;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        include fastcgi_params;
    }

    location = /favicon.ico {
        access_log off;
        log_not_found off;
        expires max;
    }

    location = /robots.txt {
        access_log off;
        log_not_found off;
    }

    error_page 404 /404.html;

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
        root /usr/share/nginx/html;
    }

    # Deny Access to Hidden Files such as .htaccess

    location ~ /\. {
        deny all;
        access_log off;
        log_not_found off;
    }

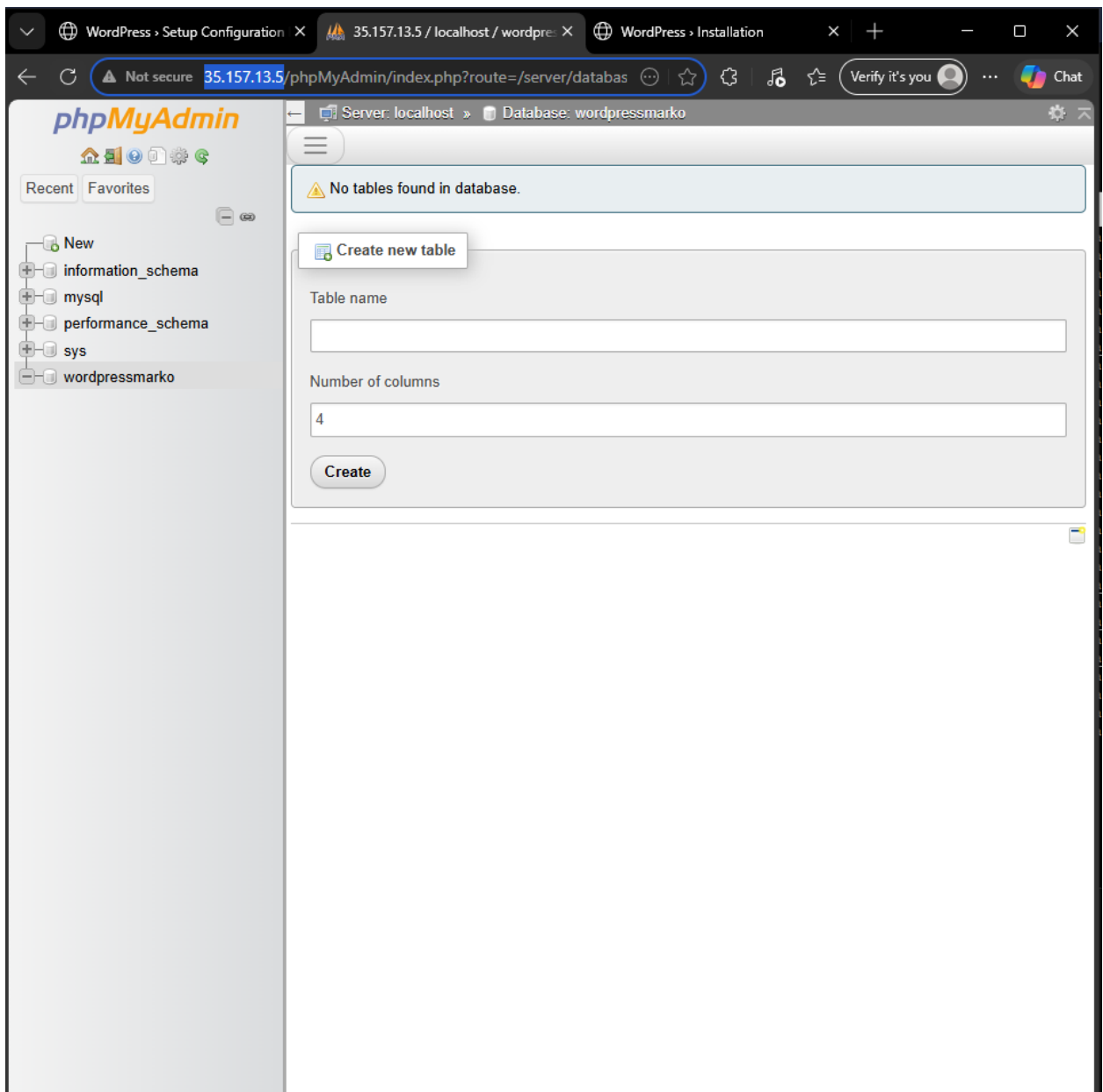
```

```

# Disallow PHP files In the WordPress uploads directory

location /wp-content/uploads/ {
    location ~ \.php$ {
        deny all;
    }
}

```



WordPress › Setup Configuration

35.157.13.5 / localhost / wordpre

WordPress › Installation

Not secure 35.157.13.5/wp-admin/install.php?language=en_US

Verify it's you

Chat



Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Names can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password

aP%PZA#Da%DxxPc^Zq

Hide

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email

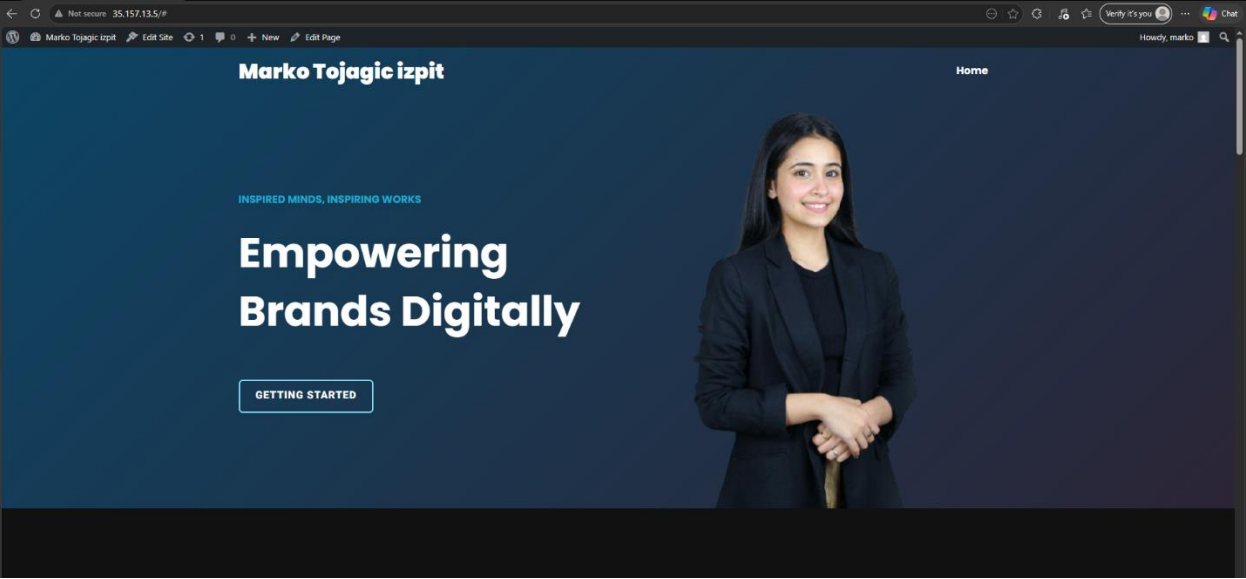
Double-check your email address before continuing.

Search engine visibility

☐ Discourage search engines from indexing this site

It is up to search engines to honor this request.

Install WordPress



Viri

1. <https://alexshomenetwork.com/installing-wordpress-on-debian-13/>
2. <https://aws.amazon.com>
3. <https://wordpress.org/>